

CLASS 01 / AUG 25 / U01

ARC 3133

GRAPHIC STANDARDS MANUAL

HOW THE WORK WILL LOOK, BEFORE THERE IS ANY WORK

NOTHING DUE

GSM ISSUED TODAY

WHAT THIS COURSE IS

Computational methods for architectural representation and digital fabrication.

WHAT YOU WILL DO

Set your own graphic standards.
Describe geometry with parameters.
Repeat it, and vary it.
Work with sampled and volumetric data.
Turn all of it into drawings and physical parts.

WHAT CARRIES THROUGH

Visual quality stays central all semester — composition, hierarchy, line weight, colour, rendering, photography.

A working graph supports those decisions. It does not replace them.

EACH PROJECT CARRIES ITS GEOMETRY INTO THE NEXT. YOU NEVER START FROM AN UNRELATED OBJECT.

THREE PARADIGMS

Each one a strict extension of the one before it.

01 PROCEDURAL

Form as a fixed sequence of operations.
You define the steps.

Output is expected.

02 PARAMETRIC

Relationships between inputs and form.
You define the rules of variation.

Output is a family — you know the range, not
each member.

03 GENERATIVE

A rule applied to its own output,
repeatedly. You define the rule and the
starting state.

Output is unknown, within a spectrum you
set.

A procedure is a fixed recipe. A parametric system is a recipe with dials. A generative system reads its own output and decides what to do next.

FOUR PROJECTS, NINE ASSIGNMENTS

Each assignment is graded once. Each project carries its geometry into the next.

P1	GRAPHIC STANDARDS MANUAL GSM four-page manual · due W10	7%
P2	CONSTRUCTIVE SOLID GEOMETRY P2a CSG process sheet · due W4 P2b 3D printed object · due W10	14%
P3	PANELING P3a module sheet · due W6 P3b arrays, two sheets · due W8 P3c plotter field · due W10 P3d site analysis · due W10	28%
P4	VOLUMETRIC DATA P4a SDF volume · due W12 P4b laser-cut slices · due FINAL	28%
MIDTERM REVIEW OF P1–P3 · WEEK 10 — OCT 27 · 7% FINAL BOOKLET · DUE FINAL · 11% PARTICIPATION · 5%		

Dates after the midterm are provisional. The course site is the reference for every due date.

HOW A CLASS RUNS

The same four things, every week.

01

REVIEW OF WORK

We open the files. Bring the editable file, not a screenshot of it.

02

DEMONSTRATION

A short tutorial or discussion of the week's unit.

03

GUIDED PRACTICE

You build the small exercise while help is in the room.

04

WORKING CRITIQUE

We look at what is on screen and say what it needs.

YOU WILL BE ASKED TO CHANGE ONE PARAMETER AND PREDICT WHAT HAPPENS. FROM THE FIRST REVIEW.

WHY THIS COMES FIRST

Every sheet you make for the next fifteen weeks uses the system you build now.

A DRAWING IS READ BEFORE IT IS UNDERSTOOD

Hierarchy is what makes an architectural drawing legible — not detail, and not effort.

Line weight, spacing and restraint do more work than anything you will model this semester. A good system makes weak work readable. A bad one hides good work.

WHAT A SYSTEM IS

Page size and margins.

A column grid.

A type hierarchy — four levels at most.

A line-weight set — four weights at most.

Three colours — black and paper count as two.

Three typefaces is the ceiling. Two is better.

FOUR PAGES. DECIDE IT ONCE, APPLY IT EVERY WEEK, REVISE IT AFTER CRITIQUE. NOT REMADE PER SHEET.

THREE MANUALS TO STUDY

Each one is a set of rules, drawn out, and applied to real cases. Read them as systems.

NYCTA · 1970

GRAPHICS STANDARDS MANUAL

A transit signage system.

Find one rule.

Find the drawing that dimensions it.

Find where it is applied.

NASA · 1975

GRAPHICS STANDARDS MANUAL

An agency identity manual.

Find one rule.

Find the drawing that dimensions it.

Find where it is applied.

MUNICH OLYMPICS · 1972

DESIGN GUIDELINES

An Olympic identity system.

Find one rule.

Find the drawing that dimensions it.

Find where it is applied.

YOUR TASK: PICK ONE RULE FROM ONE MANUAL. ADOPT IT IN YOUR GSM, AND NAME IT ON THE PAGE WHERE YOU USE IT.

All three are linked in full under Reading and documentation. Notice how much of each one is rules rather than pictures. Yours is four pages; the principle is the same.

A VISUAL IDENTITY IS A PARAMETER SET

A short list of values, decided once. Every sheet this semester reads them.

```
PAGES      = 4                # the whole manual
SHEET      = (17, 11)         # inches
MARGIN     = 0.75
COLUMNS   = 6

TYPEFACES  = 2                # 3 is the ceiling
TYPE       = {"title": 24, "heading": 14, "body": 9}
WEIGHTS    = [0.13, 0.35, 0.70] # thin, medium, heavy
COLORS     = 3                # the black counts as one
PALETTE    = ["#000000", "#FFDF7", "#C6F035"]
```

```
# every sheet this semester reads these values
```

WHAT IT MAPS ONTO

composition
grid
hierarchy
typography
line weight
palette

A BINDER OF RULES AND THIS SHORT LIST ARE THE SAME DOCUMENT, AT DIFFERENT SCALES.

ISSUED TODAY

GSM — GRAPHIC STANDARDS MANUAL

DUE WEEK 10 — Oct 27 / 7%

Build the graphic system you will communicate every later project with. It is the only assignment in the course that does not require a computational graph — it is judged purely on whether the system works and whether you used it.

DECIDE

THE SYSTEM

Grid, margins, type, line weights, colour.
Written down, not improvised.

TEST

ON REAL WORK

A diagram and a rendered image, made with
the rules you just set.

REVISE

ALL SEMESTER

Due at the midterm, revised again for the
booklet.

REQUIREMENTS — GSM

Assessed on visual quality and hierarchy 40%, consistency and usability 30%, tested applications 20%, completeness 10%.

01

Make exactly four pages. Show each graphic rule beside a tested application.

02

Use one page each for: (1) sheet size, margins, grid, columns and placement; (2) typography and hierarchy; (3) color and line weights; (4) annotations, captions and image treatment on a course sheet.

03

Use at most three typefaces, three colors including black and paper, four hierarchy levels and four line weights.

04

Study the NYCTA, NASA and Munich 1972 manuals linked under Reading and documentation. Name one rule you adopted.

05

Apply the manual to every sheet and revise it after critique.

CHECKPOINT · WEEK 2 — Sep 1 — Bring a sample sheet for an ungraded identity check.

NOW — YOUR FIRST SHEET

Rest of the class: set the page up and put something on it.



Page size set to 17 × 11 inches, with margins and a column grid



Type hierarchy chosen — up to four levels, and a reason for each



A line-weight set, from hairline to heaviest — four weights at most



Three colours you can defend, black and paper included, that survive printing



One sample sheet, with one decision about hierarchy you can explain

LEAVE WITH A SHEET. IT DOES NOT HAVE TO BE RIGHT YET.

BEFORE NEXT CLASS

WATCH

U02 — Constructive Solid Geometry. Nodes, parameters, transformations and the three Boolean operations. Watch the CSG video tutorial (Blender or Rhino + Grasshopper) linked on the Class 02 page.

INSTALL

Blender, and a project folder you can open from a clean machine. File organization is part of the workflow, not an afterthought.

BRING

Your sample sheet and the file it came from. The sheet gets an ungraded identity check next class (WEEK 2 — Sep 1).

PLAN

GSM is due at the midterm, WEEK 10 — Oct 27. Study the three manuals and pick your rule before you lay out page 1.